Ogbonna Arinze

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PROFESSIONAL SUMMARY

Fullstack engineer with 8 years of professional experience building user-facing products, backend services, and workflow-heavy business systems across healthcare, fintech, civic tech, and enterprise software. Strong experience with TypeScript, NodeJS, React, Next.js, REST APIs, CI/CD, and end-to-end product ownership, with a track record of shipping maintainable systems across frontend, backend, deployment, and operational concerns. Comfortable designing integrations, improving delivery pipelines, and translating complex domain requirements into reliable software used by internal teams and end users.

CORE COMPETENCIES

- **Fullstack Product Engineering:** TypeScript, NodeJS, React, Next.js, REST APIs, authentication flows, and workflow-heavy application development.
- **Frontend Delivery:** React, Next.js, Redux, React Query, Tailwind CSS, reusable UI systems, responsive layouts, and user-facing product interfaces.
- **Backend and Data:** NestJS, NodeJS services, PostgreSQL, MongoDB, Prisma, relational and NoSQL data modeling, and API integrations.
- **Infrastructure and Operations:** Docker, Docker Compose, AWS, GitHub Actions, CI/CD pipelines, health checks, deployment workflows, and production ownership.
- **Quality and Performance:** Playwright, Jest, Lighthouse, Chrome DevTools, code splitting, caching, and test-driven delivery in CI environments.
- **Domain and Team Impact:** Fintech workflows, operational platforms, internal tools, cross-functional collaboration, mentorship, and engineering standards.

PROFESSIONAL EXPERIENCE

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered budget analysis and corruption tracking platform with 700+ documents, 708K+ vector embeddings, and 37-state coverage.

- Architected and built the entire platform as an Nx monorepo managing 6 applications: NestJS API backend, Next.js citizen-facing web app, Next.js admin dashboard, NestJS ingestion pipeline, Next.js marketing site, and a Remotion video generation service.
- Designed and implemented a multi-agent AI system using Mastra with 5 specialized agents for intent classification and domain-specific RAG-powered analysis.
- Built a hybrid RAG pipeline combining pgvector similarity search with BM25 full-text search, merged via Reciprocal Rank Fusion and Cohere reranking across 4 vector indexes.
- Designed the PostgreSQL 16 schema with Prisma v7, including pgvector extension support and custom migration tooling to handle drift detection correctly.
- Configured CI/CD with GitHub Actions, Playwright end-to-end tests across 4 projects, and Docker Compose production deployment with multi-stage Dockerfiles, health checks, memory limits, and graceful shutdown handling.

Genentech (Roche) San Francisco, CA (Remote)
Frontend Engineer *May 2023 - Present*

- Spearheaded the redesign of the main product for connecting and transferring clinical trial site data to various upstream EHR systems, migrating from a Vue.js web application to a Chrome extension in React.
- Maintained and extended the existing Vue.js codebase during the migration period, fixing bugs and shipping features in parallel with the React rebuild.
- Implemented authentication and authorization workflows on the frontend using JWT-based flows, HTTP-only cookies, protected routes, secure token refresh handling, and permission-based UI rendering in collaboration with backend engineers.
- Introduced Redux for state management and React Query for request caching, refactoring legacy React code from class components to functional components and improving maintainability across the application.
- Optimized application performance by implementing lazy loading, code splitting, image optimization, responsive layouts, memoization, and efficient asset delivery via CDN.
- Increased stakeholder confidence by writing end-to-end tests with Playwright in the CI/CD pipeline, publishing Allure reports on GitHub Pages, and sending Slack notifications with links to generated test results on every push.
- Co-developed a full-stack Next.js chat widget powered by a Retrieval-Augmented Generation system using OpenAI GPT models, LangChain, Vercel AI SDK, and Chroma vector database, enabling users to connect data sources and query them in real time.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Played a pivotal role in a team of 3 frontend engineers to build a high-volume bank Settlement Solution, significantly enhancing transaction processing capabilities to handle thousands to millions of transactions daily, boosting transaction settlement rates by 80%.
- Spearheaded the development of critical modules for the Settlement Solution, including MasterCard, VISA, Express-cash, and E-commerce modules, using React with Redux for state management, RxJS for data streaming, and Axios for asynchronous API requests.
- Developed a real-time account reconciliation solution, boosting the reconciliation of daily transactions by 60% through innovative transaction processing and data management strategies.
- Enhanced system performance and user experience by adopting the native browser EventSource API for real-time data streaming, replacing conventional socket.io methods for faster data updates and a more responsive interface.
- Built dynamic data visualization dashboards using React and react-chartjs, streamlining management of user roles and permissions with Redux Toolkit and Ant Design UI.

Founders Lair Beijing, China (Remote)

Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Translated business requirements into efficient database models and schemas, collaborating closely with the business team to ensure technical solutions met business needs.
- Developed NodeJS RESTful APIs to power new features, focusing on testable, maintainable, and well-documented code.
- Transformed user journeys and Figma designs into reusable React components, ensuring a consistent, intuitive user interface.
- Engineered a CI/CD pipeline for automated deployment of APIs to testing environments on Heroku and production servers on AWS, while guiding two interns through frontend and backend projects.

Descasio Lagos, Nigeria (Hybrid)

Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive user interface using React, achieving a 25% boost in user engagement while collaborating closely with software architects to create a scalable frontend architecture and accelerate development by 30%.
- Enhanced the development workflow by introducing automated testing frameworks, including Jest and Enzyme, which led to a 20% decrease in bug reports.
- Utilized Storybook for documenting React components, streamlining change management and ensuring clear communication of development processes while supporting user story planning, estimates, and work plans.
- Optimized frontend performance and load efficiency by identifying and resolving render bottlenecks using React Profiler and Lighthouse Audits, reducing bundle size by 30-40%, which led to a 35% faster load time and smoother runtime experience across devices.

SureTrade Business Solutions Nigeria

Frontend Developer *Mar 2018 - Oct 2019*

- Designed MongoDB database schemas and implemented them in NodeJS for multiple applications.
- Worked with a team of 5 to develop SureEdu, a school management portal for school owners to manage staff, students, and guardians.
- Responsible for server-side web application logic as well as integration of the frontend, developing user use cases, converting them into database schemas and models, and testing API endpoints.

EDUCATION

University of Nigeria Nsukka, Enugu

BSc Computer Science, 4.1 CGPA *Sep 2019*